INCH-POUND

MIL-I-27191C(USAF)
15 March 1995
SUPERSEDING
MIL-I-27191B(USAF)
1 November 1979

MILITARY SPECIFICATION

INDICATOR, POSITION, VARIABLE NOZZLE

This specification is approved for use by the Department of the Air Force and is available for use by all Department and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements for two types of 28V DC, integrally lighted, variable nozzle position indicators.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-B-601	Boxes Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 6620

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MILITARY

MIL-P-116	Preservation, Methods Of
MIL-T-5583	Transmitter, Position, 28-Volt DC
MIL-L-6363/8	Lamps, Incandescent, Aircraft Use, Single Contract Bayonet Candelabra, Base, Reflector Type
MIL-I-7062	Indicators, Position, Control Surfaces, 28 Volt DC, General Specification For
MIL-T-26609	Transmitter, Position
MIL-L-27160	Lighting, Instrument, Integral, white, General Specification For

STANDARDS

FEDERAL

FED-STD-595	Colors

MILITARY

MIL-STD-129	Marking For Shipment And Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-810	Environmental Test Methods and Engineering Guidelines Packaging Requirements
MS24515	Lamp, Sub Miniature
MS24550	Transmitter, Position
MS33585	Pointers, Dial, Standard Design Of Aircraft Instrument

AIR FORCE-NAVY AERONAUTICAL

AN5786	Transmitter - Position, 28 Volt DC
AN5787	Transmitter - Position, Gearless

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

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American Society For Testing And Materials (ASTM)

ASTM D3951	Packaging, Commercial
ASTM D4727	Standard Specification For Corrugated And Solid Fiberboard Sheet Stock (Container Grade) And Cut Shapes

(Application for copies should be addressed to: ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Non-Government Standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Qualification. The indicator furnished under this specification shall be a product which has been tested and passed the qualification test specified herein, and has been listed on or approved for listing on the applicable qualified products list (see 4.4 and 6.3).

3.2 General Specification. The requirements of MIL-I-7062 apply as requirements of this specification with the exceptions and deletions called out herein. When the two specifications conflict, this specification shall govern.

3.3 Design and construction. The indicator shall be designed for operation with position transmitters in accordance with MIL-T-5583 and MIL-T-26609. The pointer shall travel 245 mechanical degrees for full scale operation 0 to 100 percent dial indication. The pointer shall indicate linear travel of 180 mechanical degrees per 90 degrees rotation of an AN5786 or MS24550-1 transmitter and 180 degrees rotation of an AN5787 or MS24550-2 transmitter.

3.3.1 Resistance. The direct-current resistance of the indicator measured between the connector pins A, B, and C shall be 475 ±100 ohms. Deviation from the average of the resistance values between the three coils shall not exceed ±1-1/2 percent.

3.4 Case. The case shall be as shown in MS24752. The case shall provide a hermetically sealed enclosure for the internal mechanism of the indicator.

3.4.1 Case design. The design of the indicator case shall be such that the internal mechanism may be removed from the case, replaced, and the case resealed. This shall be accomplished without the use of special tools and fixtures, unless they are approved by the procuring activity. The hermetic sealing shall be so accomplished that the seal will not be dependent upon materials which will be affected by the action of any atmosphere to which the indicator may be subjected.

3.4.1.1 Filler tube. A copper-alloy tube having an inside diameter of 0.06 inch and an external length of approximately one inch shall be provided on the rear of the indicator case. It shall be recessed beneath the end of the case. A tapered filler plug may be used in lieu of the filler tube. The tube or plug shall be used for evacuating or filling the case, as required.

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3.4.1.2 Fogging. There shall be no evidence of moisture or oil fogging of the cover glass when the indicator is subjected to a temperature of 85° ±5° C for a minimum of two hours and then the face of the glass immediately immersed in an ice water bath for three minutes.

3.4.2 Filling medium. The filling medium shall be of at least 98 percent purity, free of dust particles, and shall contain not more than 0.006 milligram of water vapor per liter (dew point - 65° C) at the filling pressure. The filling medium shall be 88 to 92 percent nitrogen and the remainder helium. Other concentrations of helium shall not be used except with prior approval of the procuring activity. The absolute pressure of the filling medium in the case shall be approximately one atmosphere.

3.4.3 Cover glass position. The distance between the outer surface of the cover glass and the front edge of the case shall be the minimum practicable and shall not exceed 1/32 inch.

3.5. Dial. The dial shall be as shown in MS24572. The dial shall be securely fastened to the frame of the mechanism in such a manner that it will not loosen or turn when the indicator is vibrated. The diameter of the dial shall be not lest the 1-3/4 inches when measured across the outside ends of the graduations.

3.5.1 Markings. The finish and dimensions of the dial markings shall be as follows:

Markings	Height Or Length inches +0.016	Width of Line or Graduatio Inches + 0.005	Color FED-STD-595
Numerals 0, 20, 40, 60, 80, 100	0.125	0.016	No. 37875
Major graduations	0.141	0.025	No. 37875
Minor graduations	0.093	0.016	No. 37875
Lettering "NOZZLE"	0.125	0.020	No. 37875
Lettering "PERCENT"	0.0625	0.010	No. 37875
Lettering "OPEN"	0.093	0.016	No. 37875
Lettering "MIN"	0.093	0.016	No. 37875
Lettering "MAX"	0.093	0.016	No. 37875
Lettering "MS24572" (3 or 4 as applicable)	0.093	0.016	No. 37038

3.6 Pointer. The pointer shall conform essentially to MS33585-8 and shall be suitable for integral lighting. The pointer hub shall not obscure the marking "Percent" on the dial.

3.6.1 Pointer dampening. The pointer shall be sufficiently dampened to prevent an oscillations in excess of ± 1.0 percent while operating and while the transmitter is rotated in short rapid increments. The pointer shall dampen to zero motion within a time period of 1.0 ± 0.25 second after it is given a step increment of 65 percent.

3.7 Electrical connector. Unless otherwise specified, the electrical connector shall be a Bendix Type PT-IH-10-10-6P or equal.

3.8 Integral lighting. The indicator covered by this specification shall be integrally lighted in accordance with the specifications indicated below except as otherwise specified:

MS24572-4 -MIL-L-27160

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3.8.1 Lamp. The light source for the indicator shall be three or more MIL-L-6363/8 lamps.

3.8.1.1 Lamp position. The lamps shall be located within the hermetically sealed enclosure.

3.8.1.2 Lamp replacement. The lamp installation shall be designed so that replacement lamps can be accurately positioned when replacement is necessary. Replacement will be accomplished at overhaul depots.

3.9 External wiring. The external wiring shall be a shown on figure 1.

3.10 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.3)

b. Quality conformance inspection (see 4.4).

4.3 Inspection conditions.

4.3.1 For the purpose of this specification the identification of the tests shall be as follows:

MIL-I-27191	MIL-I-7062
Acceptance tests	Inspection tests
Sampling plan A	Sampling tests
Sampling plan B	Special tests
Qualification tests	Qualification tests

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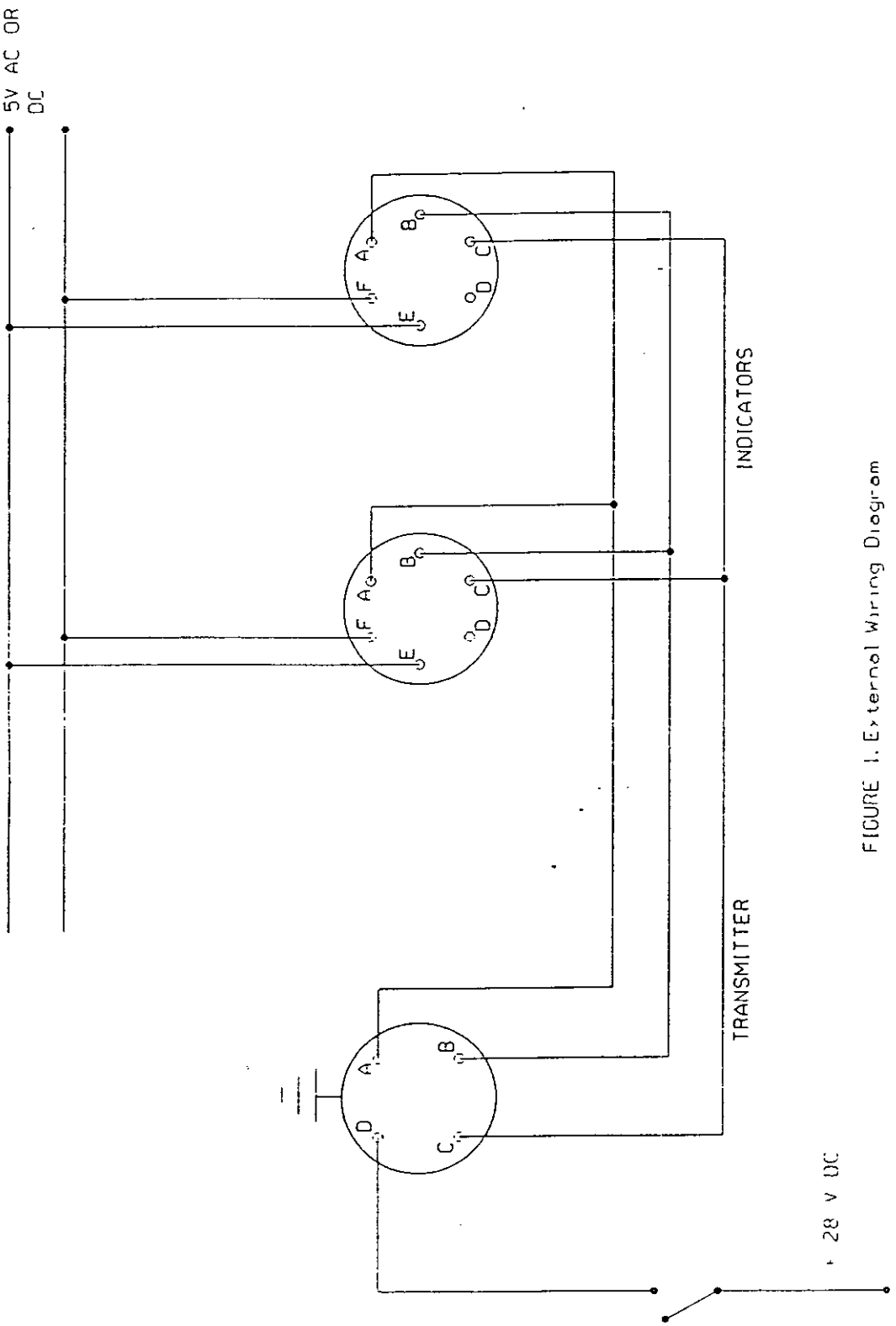


FIGURE 1. External Wiring Diagram

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4.3.2 Scale error test. The scale error shall be read from the test transmitter scale with the indicator pointer on each graduation of the indicator dial.

4.3.2.1 A scale error test shall be conducted using two indicators with one master transmitter. The accuracy of each indicator during this test shall not exceed 1.25 times the single indicator accuracy specified.

4.3.2.2 Accuracy. The accuracy for a single indicator shall be as follows:

a. ± five mechanical degrees for room and high temperature tests

b. ± six mechanical degrees for vibration and low temperature tests.

4.4 Quality conformance inspection. Quality conformance inspections shall consist of individual tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.1 Individual tests. The indicator shall be subjected to and meet the individual tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.2 Sampling plan A.

4.4.2.1 The indictor shall be subjected to and meet the sampling plan A tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.2.2 Vibration error. The indicator shall be mounted on vibration stand in its normal operating position. The indicator shall be electrically connected for operational and lighting conditions. The indicator shall be subjected to vibration at an amplitude setting between 0.009 and 0.011 inch at frequencies varying from 300 to 3,000 cpm. During vibration maximum amplitude of oscillation and variation of the movable pointer shall not exceed 1/32 inch. All of the lamps shall be operating at the completion of this test.

4.4.2.3 Pointer dampening. The indictor shall be electrically connected for operation, and the transmitter shall be rotated in short rapid increments. The amount of pointer oscillation shall not exceed ± 1.0 percent. The indictor shall then be connected to two transmitters which shall be electrically separated by a switch. One transmitter shall be set to read 20 percent and the other transmitter shall be set to read 85 percent. The indicator connections shall be switched from one transmitter to the other (by use of the switch). The time interval from initiation of the switch to the instant the pointer comes to a complete stop shall be measured and shall not exceed 1.0 ± 0.25 second. A stopwatch may be used to measure the time interval.

4.4.3 Sampling plan B.

4.4.3.1 The indicator shall be subjected to and meet the sampling plan B tests specified in MIL-L-25467 or MIL-L-27160, as applicable.

4.4.3.2 Vibration failure. The indicator shall be subjected to the vibration test of MIL-STD-810, method 514.2, procedure I. The test shall follow Curve C of Figure 514.2-2 to a maximum frequency of 52 Hz. The test shall be conducted while the indicator is being operate don 28V DC and while the bulbs are illuminated with 5V power.

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4.4.3.3 Fogging. The indicator shall be placed in a controlled ambient temperature of 85 ± 5° C for a minimum of two hours. Within ten seconds after removal from the oven, the indicator face shall be immersed to a depth of 1/4 to 1/3 inch in an ice water bath maintained at 0° to 5° C. After three minutes, the indicator shall be removed from the ice bath and quickly wiped dry (compressed air shall not b used). The cover glass and dial shall immediately be inspected for evidence of fogging or damage. Evidence of fogging or damage shall be cause for rejection.

4.5 Qualification testing.

4.5.1 Qualification test samples. The qualification test samples shall consist of three indicators, samples shall be identified with the manufacturer's own part number and nay additional information required by the procuring activity.

4.5.2 Qualification tests. The qualification tests shall consist of all the tests specified herein.

4.6 Packaging inspection.

4.6.1 Quality conformance. The inspection of the preservation-packaging and interior package marking shall be in accordance with group A and B quality conformance inspection requirements, Section 4 of MIL-P-116. The sampling and inspection of the packing and marking for shipment and storage shall be in accordance with the quality assurance provisions of applicable container specification and the marking requirements of MIL-STD-129.

4.6.2 First article. When the unit container is capable of serving as the shipping container first article inspection and rough handling tests as outlined in Section 4 of MIL-P-116, shall be accomplished followed by a functional test of the unit to ensure freedom from operational malfunction.

5. PACKAGING

5.1 Preservation-packaging. Preservation shall be level "A," "C," or "Industrial" as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Indicator shall be cleaned in accordance with process C-1 of MIL-P-116.

5.1.1.2 Drying. Indicator shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation-application. Not applicable.

5.1.1.4 Unit packaging. Unless otherwise specified by the contracting activity, each indicator shall be packaged in quantity unit packs of one each in accordance with Method 1A8 of MIL-P-116. Each indicator shall be packed in fiberboard shipping containers conforming to PPP-B-636, "Box, Shipping, Fiberboard" or ASTM D4727, "Standard Specification For Corrugated And Solid Fiberboard Sheet Stock (Container Grade) And Cut Shapes." Sufficient cushioning material between the bag and unit container shall be provided. The cushioning material shall be of the proper type, density, and thickness to ensure shock transmission does not exceed peak values in G's established for the compass when completed packs are subject to the rough handling drop test of MIL-P-116.

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5.1.2 Level C. Each indicator shall be clean, dry, and packaged in a manner that will afford adequate protection against corrosion, deterioration, and physical damage during shipment from the supply source to the first receiving activity.

5.1.3 Industrial. The industrial preservation of indicator shall be in accordance with ASTM D3951.

5.2 Packing. Packing shall be level "A," "B," "C," or "Industrial" as specified (see 6.2).

5.2.1 Level A. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-601, Styles A or B, Class overseas, unless otherwise specified by the contracting activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, minimum tare and cube consistent with the protection required.

5.2.2 Level B. Indicator packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636 or ASTM D4727, unless otherwise specified by the contracting activity. All other requirements, as specified by the contracting activity, shall apply.

5.2.3 Level C. Packing shall be applied which affords protection from domestic shipment from the supply source to the first receiving activity for immediate use. This level shall conform to applicable carrier rules and regulations.

5.2.4 Industrial. The packaged indicator shall be packed in accordance with ASTM D3951.

5.3 Marking. In addition to any other markings required by the contract or order (see 6.2), interior and exterior containers shall be marked in accordance with MIL-STD-129.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The variable nozzle position indicator covered by this specification is intended to be used for indicating the position of variable exhaust nozzles, on turbo jet engines so equipped, in percent of maximum area. With this indication, the pilot is able, with proper interpretation, to infer the performance of the engines afterburner complex.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODIIS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Levels of preservation, packaging, and packing required.

d. Whether sampling plan B tests are to be conducted.

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6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for opening of bids, qualified for inclusion in Qualified Products List (QPL No.) whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or purchase orders for the products covered by this specification. The activity responsible for the Qualified Products List is Oklahoma City Air Logistics Center/TICLA, Tinker AFB OK 73145-5990 and information pertaining to qualification of products may be obtained from that activity.

6.4 Subject term (key words) listing. Recommended key words are:

Hermetically sealed
Integrally lighted
Nozzle position indicator

6.5 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:
Air Force - 99

Reviewer:
DGSC - GS

Preparing activity:
Air Force - 99

Project number:
6620-F535

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2. 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-27194C(USAF)	95/03/15

3. DOCUMENT TITLE
Indicator, Position, Variable Nozzle

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
OC-ALC/TICLA		336-5460
c. ADDRESS (Include Zip Code) 3001 Staff Dr, Tinker AFB OK 73145-5990	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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